Feedback Frameworks & Large Language Models

AI Research Paper Explainer

Introduction

In the research paper *From First Draft to Final Insight: A Multi-Agent Approach for Feedback Generation*, the authors propose a novel method for improving feedback in educational settings (Cao et al., 2025). They introduce a multi-agent generation-evaluation-regeneration (G-E-RG) framework, using large language models (LLMs) to automatically generate feedback, evaluate its quality, and then regenerate or refine it. Feedback is a cornerstone of the learning process, enabling students to reflect on their efforts and identify areas for improvement. It is through feedback that the core learning objectives of a course or learning material are internalized and deeply understood. However, not all feedback meets these standards, as its effectiveness depends on factors such as timing, format, and most importantly, the quality of the content itself. High-quality feedback does more than provide information; it is grounded in sound educational theory and tailored by teachers to the abilities of each student.

With the rise of Large Language Models in recent years, interest has grown in their potential to support feedback delivery and help teachers maintain consistent quality across large student groups.This interest, unfortunately, has not materialized into a classroom-ready system yet due to the limitations LLMs still exhibit. LLMs often generate verbose responses regardless of their actual knowledge of the subject, which can result in fabricated details that may inadvertently mislead students. Another limitation is the absence of pedagogical grounding; LLM responses are not informed by established educational theories or feedback frameworks. To address these issues, researchers have developed techniques such as Retrieval-Augmented Generation (RAG) and Chain-of-Thought (CoT) prompting, which have shown considerable success in reducing hallucinations.

Still, the challenge of embedding a robust educational framework into LLMs remains unresolved. To address this, the authors introduce the Generate–Evaluate–Regenerate (G-E-RG) system, a multi-agent approach with role assignments. In this framework, initial feedback is produced using various prompt-engineering techniques and feedback frameworks, then automatically evaluated and refined.

Research Questions

The authors propose three research questions:

1. Which combination of prompt techniques and feedback framework generates the initial highest-quality feedback?
2. How accurately can LLMs evaluate the generated feedback?
3. To what extent does the "generation, evaluation, and regeneration" method improve the quality of LLM-generated feedback?

Methodology

The dataset was collected from the Fall 2024 graduate course E-Learning Design Principles and Methods at Carnegie Mellon University, which contains specialized educational knowledge unlikely to appear in LLM pretraining. It included 63 multiple-choice questions (152 responses) and one open-ended question (56 responses), totaling 208 responses.

The G-E-RG pipeline consists of three GPT-4o agents and a course-slide database. The first agent generates feedback using two prompting strategies—Zero-shot and RAG_CoT—and three frameworks: no framework, a learner-centered model (Ryan et al., 2021), and a knowledge-transmission model (Hattie & Timperley, 2007). In RAG_CoT, slides are retrieved via semantic similarity for use in feedback.

The second agent evaluates feedback with a rubric covering four dimensions:

Feedback Evaluation Rubric		
Dimension	**Sub-indicator**	**Value**
Reliability	Evaluation accuracy (EA) Retrieved Slide Accuracy (RSA)	True/False
Components	**C1-critiques**: Comments that provide critiques about performance **C2-strengths:** Comments that highlight strengths of performance **C3-actionable**: Comments that provide actionable information for future performance **C4-agency**: Comments that encourage learner agency	Yes/No
Features	**F1-positive:** Comments encourage a positive learner effect. **F2-usable:** Comments are usable for the learner **F3-relationship:** Comments strengthen teacher and learner relationships **F4-dialogue:** Comments invite dialogue about feedback **F5-independence:** Comments promote learner independence	1 (No/Minimal) 2 (Moderate) 3 (Strong)
Simplicity	The generated feedback's word count	0-N

Takeaways

First-Round Feedback

RAG_CoT prompting outperformed the baseline across all metrics. RAG_CoT_knowledge achieved the highest evaluation accuracy (≈93%) and strong feature scores (positive, usable, relationship), while the learner framework produced more dialogic and independence-supporting feedback. Coverage for critiques (C1) was low across methods, but RAG_CoT_learner achieved the most balanced performance. A tradeoff emerged: framework-based feedback was longer (up to 174 words), underscoring the need for regeneration to condense outputs.

Automatic Evaluation Accuracy

LLMs demonstrated competence as evaluators of reliability and concrete feedback elements, with F1 scores above 0.9 for strengths and actionable suggestions. Performance dropped for nuanced pedagogical features such as usability (≈0.75) and independence (≈0.73), suggesting a continued need for human calibration.

Second-Round Feedback

The second round significantly improved evaluation accuracy (up to +13%), with RAG_CoT_learner reaching 97.6%. Coverage of all four components rose to nearly 100%, and feature scores improved by 0.5 to 1 points on average. Regeneration also reduced verbosity across all methods, making feedback clearer and more concise. Remaining challenges lie in refining subtle features like relationship-building and independence.

Application

The G-E-RG pipeline illustrates an emerging trend in the EdTech market that may not be dominated by major AI industry players such as OpenAI or Google, as some might have expected. This is partly due to ongoing ethical considerations related to data collection: educational institutions may be reluctant to share sensitive student data, and partly because there is no single accepted philosophy for feedback frameworks in education. As a result, systems like G-E-RG can offer locally hosted solutions tailored to the pedagogical approaches of individual schools and educators. Compared with more generalized systems such as ChatGPT's "Study Mode," which does not explicitly state its theoretical underpinnings, these approaches can provide feedback that is more closely aligned with specific teaching philosophies. In this way, teachers retain greater control over the design and delivery of instruction and feedback, supporting a more customized and context-sensitive learning experience.

In addition, systems of this kind can help ease some of the workload placed on teachers. Research has shown that many teachers experience stress, burnout, and other mental health challenges, with classroom size identified as one contributing factor. These challenges can influence not only teachers' well-being but also the quality of their instruction. By integrating supportive technologies into classrooms, teachers may be able to manage their responsibilities more effectively, which in turn can contribute to a more consistent and sustainable learning environment for students.

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